

## Counting solutions to equations in finite groups

Let  $G$  be a finite group,  $f: G \rightarrow \mathbb{C}$ . Let  $F_n$  denote the free group in  $n$  generators. Let  $w \in F_n$ . Define

$$\{f, w\} = \frac{1}{|G|^n} \sum_{x \in G^n} f(w(x)).$$

Lemma: Let  $\chi: G \rightarrow \mathbb{C}$  be an irr. character. Then  $\forall z \in G$ ,

$$\frac{1}{|G|^n} \sum_{x \in G^n} \chi(w(x)z) = \{ \chi, w \} \frac{\chi(z)}{\chi(1)}.$$

This is clear if  $\chi$  is multiplicative. For  $w=x$ , it says

$$\frac{1}{|G|} \sum_{x \in G} \chi(xz) = \frac{\chi(z)}{\chi(1)} \frac{1}{|G|} \sum_{x \in G} \chi(x).$$

Proof: Let  $\rho: G \rightarrow \text{Aut}(V)$  denote the corresponding representation, and

$$W = \frac{1}{|G|^n} \sum_{x \in G^n} \rho(w(x)) \in \text{End}(V)$$

Then 
$$W = \frac{1}{|G|^n} \sum_{x \in G^n} \rho(w(zxz^{-1}))$$

$W$  is  $G$ -linear

$$= \frac{1}{|G|^n} \sum_{x \in G^n} \rho(zw(x)z^{-1}) = \rho(z)W\rho(z)^{-1}$$

Hence,  $W$  is  $G$ -linear. Since  $\rho$  is irreducible,  $W = \lambda I$  by Schur's lemma. Taking traces, we get

$$n\lambda = \text{Tr } W = \frac{1}{|G|^n} \sum \chi(w(x)) = \{ \chi, w \}.$$

Hence, 
$$W\rho = \frac{\{ \chi, w \}}{\chi(1)} \rho$$

$$\frac{1}{|G|^n} \sum \rho(w(x)).$$

Taking traces yields the result. □

Prop: Let

$$N(z) = \#\{x \in G^n \mid w(x)z = 1\}.$$

Then

$$N(z) = |G|^{n-1} \sum_{\chi \text{ irr.}} \{ \chi, w \} \chi(z).$$

Proof: Let  $\delta(x) = \begin{cases} 1 & x=1 \\ 0 & x \neq 1 \end{cases}$ . Write  $\delta = \sum_{\chi \text{ irr.}} c_{\chi} \chi$ . Then

$$c_{\chi} = (\chi, \delta) = \frac{\chi(1)}{|G|},$$

and

$$\begin{aligned} N(z) &= \sum_x \delta(w(x)z) = \sum_x \sum_{\chi \text{ irr.}} \frac{\chi(1)}{|G|} \chi(w(x)z) \\ &= |G|^n \sum_{\chi} \frac{\chi(1)}{|G|} \frac{\{ \chi, w \}}{\chi(1)} \chi(z). \end{aligned}$$

□